








Statistics of Financial Markets

Seminar 11: Fractal markets hypothesis and long memory



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); Abnormal Return
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **DFA**: Detrended Fluctuation Analysis
- **EMH**: Efficient Market Hypothesis
- **EODHD**: EOD Historical Data (market-data provider)
- **FMH**: Fractal Market Hypothesis
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GPH**: Geweke–Porter-Hudak (log-periodogram estimator)
- **MA**: Moving Average
- **OLS**: Ordinary Least Squares
- **SE**: Standard Error

Today's Question and Seminar Structure

- Question of the seminar
  - **how can we measure the long memory of a market without seeing memory where there is none?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise Map (next slide)
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	R/S of eight returns step by step; H from three points of an R/S plot	Solved, Proposed	A1
A3, A4	fractional Gaussian noise: autocorrelations, 10-day risk, VaR 1% (value at risk)	Solved, Proposed	A3
A5, A6	ARFIMA(0,d,0): weights and autocorrelations against an AR(1)	Solved, Proposed	A5
B1, B2	R/S, DFA, GPH and Lo's test with Monte Carlo bands: S&P 500; BET, TLV	Solved, Proposed	B1
B3, B4	long memory in $\lvert r_t \rvert$ and r_t^2, the shuffle test: S&P 500; Bitcoin	Solved, Proposed	B3
B5, B6	rolling DFA exponents with a Monte Carlo band: S&P 500; DAX, TLV	Solved, Proposed	B5
C1, C2	does memory change in crises? what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it after the model; the solution is discussed in the seminar
- A: calculations on paper; B: real data, in the notebook, each ending with an interpretation question; C: open questions, among them the audit of an AI answer

Data Used

Series	Source	Price used	Period
S&P 500, DAX, BET	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
Banca Transilvania (TLV)	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, Chapter 2)
- In the notebook: `returns('bet')`, `hurst_rs(x)`, `hurst_dfa(x)`, `gph(x)`, `lo_test(x)`, `mc_null(N)`; no account or key is needed

Prerequisites for Today (1/15): The Fractal Market Hypothesis

- **FMH** (fractal market hypothesis) Peters (1994): a market is stable when investors with many horizons (minutes to decades) trade with each other
  - a short-horizon seller finds a long-horizon buyer, so prices stay liquid
  - in a crisis, all horizons shrink to the shortest one: no buyers, no liquidity
  - testable through scaling and memory across horizons (the next slides)
- **EMH** (efficient market hypothesis), weak form (Chapter 7)
  - past returns do not help to predict the direction of future returns
  - under the EMH, log prices behave like a random walk: independent increments

Prerequisites for Today (2/15): Self-Similarity and the Hurst Exponent

- **Self-similarity**: zooming in or out in time looks the same, after rescaling the values

$$X_{ct} \overset{d}{=} c^H X_t \quad \text{for every } c > 0$$

  - X_t: the log price at time t; $\overset{d}{=}$: the same distribution; stretching time by c stretches the values by c^H

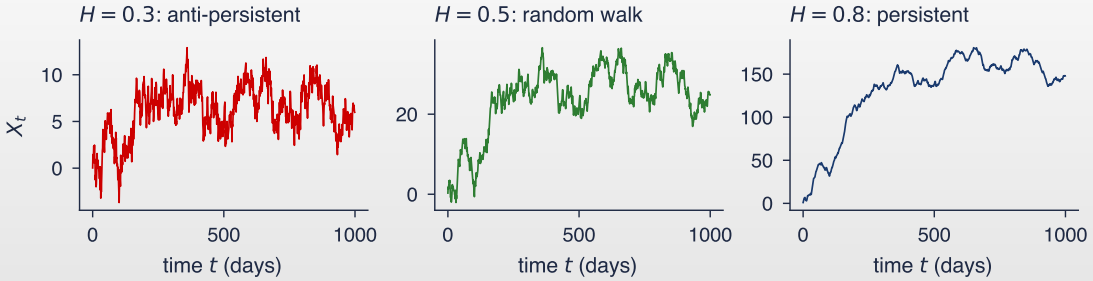
- **Hurst exponent** $H \in (0, 1)$
  - $H = 0.5$: independent increments (random walk, consistent with the EMH)
  - $H > 0.5$: **persistence**, a rise tends to be followed by a rise; $H < 0.5$: **anti-persistence**, rises and falls alternate
  - H is a scaling exponent, not a probability

- **Fractal dimension** of a price path

$$D = 2 - H$$

  - between 1 (a smooth line) and 2 (a path so rough that it fills the plane); a small H gives a rough path

Prerequisites for Today (3/15): Fractional Brownian Motion



- **Fractional Brownian motion** (fBm) Mandelbrot & Van Ness (1968): a self-similar Gaussian process with exponent H; $H = 0.5$ is the ordinary Brownian motion
- The same random numbers with three values of H: the smaller H, the rougher the path; the larger H, the smoother the trends
- **Fractional Gaussian noise** (fGn): the daily increments of fBm, i.e. the returns

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Prerequisites for Today (4/15): Scaling of Multi-Day Risk

- From self-similarity, the h-day return $X_{t+h} - X_t$ has standard deviation

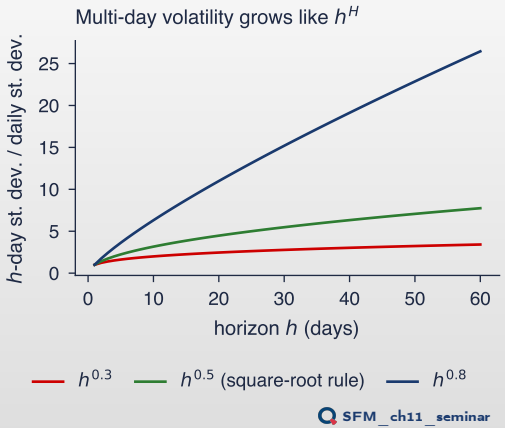
$$\sigma_h = h^H \, \sigma_{\mathbf{1}}$$

  - $\sigma_{\mathbf{1}}$: the daily standard deviation; h: the horizon in days
  - $H = 0.5$: the square-root-of-time rule $\sqrt{h}\,\sigma_{\mathbf{1}}$

- **Multi-day VaR 1%** (Normal, mean 0)

$$\mathrm{VaR}^{(h)} = -z_{\mathbf{0.01}} \, \sigma_{\mathbf{1}} \, h^H, \qquad z_{\mathbf{0.01}} = -2.326$$

  - $H > 0.5$: losses add up over days and $\sqrt{h}$ understates the risk; $H < 0.5$: it overstates it



Prerequisites for Today (5/15): The R/S Statistic

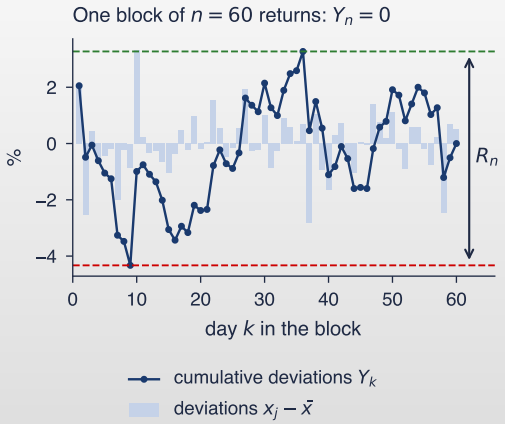
- A block $x_1, \ldots, x_n$ of n returns with mean $\bar{x}$ Hurst (1951)
  - **cumulative deviations**:

$$Y_k = \sum_{j=1}^{k} (x_j - \bar{x}), \qquad k = 1, \ldots, n$$

  - $Y_n = 0$ always, because the deviations from the mean add up to zero
- **Range** and **standard deviation**

$$R_n = \max_k Y_k - \min_k Y_k, \qquad S_n = \sqrt{\frac{1}{n} \sum_{j=1}^{n} (x_j - \bar{x})^2}$$

  - R_n: how far the cumulative sum wanders; $(R/S)_n = R_n/S_n$ has no unit
  - runs of equal signs make Y_k drift, so R_n is large; alternating signs keep it small



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Prerequisites for Today (6/15): Estimating H from R/S

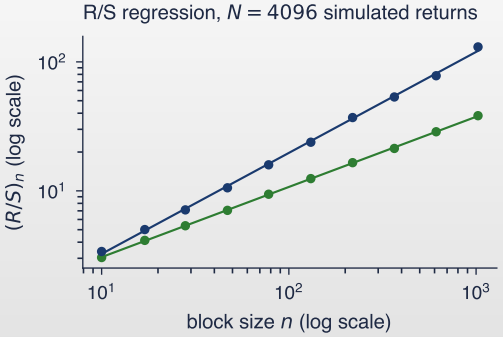
- **Power law**: $(R/S)_n \approx c\, n^H$, with c a constant; taking logs:

$$\log(R/S)_n = \log c + H \log n$$

  - average R/S over the non-overlapping blocks of each size n; regress $\log(R/S)_n$ on $\log n$; the slope is $\hat{H}$
  - three block sizes equally spaced on a log scale: the OLS slope equals the slope between the two outer points

$$\hat{H} = \frac{\log(R/S)_{n_3} - \log(R/S)_{n_1}}{\log n_3 - \log n_1}$$

- **Small-sample bias** Anis & Lloyd (1976)
  - for i.i.d. data the R/S slope is above 0.5: about 0.544 for $N = 6\,714$



- i.i.d. Normal: slope $\hat{H} = 0.54$
- fGn, $H = 0.8$: slope $\hat{H} = 0.78$

Prerequisites for Today (7/15): Long Memory and Short Memory

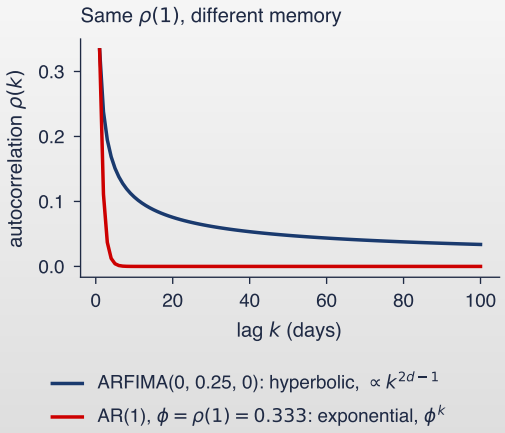
- **Long memory** (Franke, Härdle & Hafner (2019), 14.1): the autocorrelations decay hyperbolically

$$\rho(k) \sim C\, k^{2d-1}, \qquad 0 < d < 0.5$$

  - $\rho(k)$: the autocorrelation at lag k; $C > 0$: a constant; d: the memory parameter; $\sim$: for large k
  - the autocorrelations are not summable: $\sum_k \rho(k) = \infty$
- **Short memory**: an AR(1) $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$

$$\rho(k) = \phi^k$$

  - an exponential decay: after a few lags the past is forgotten
  - the chart: the same $\rho(1)$, very different $\rho(k)$ at lag 50



Prerequisites for Today (8/15): Fractional Gaussian Noise

- **fGn** with exponent H Mandelbrot & Van Ness (1968): the autocorrelation at lag k

$$\rho(k) = \frac{1}{2}\left(|k+1|^{\mathbf{2}H} - 2|k|^{\mathbf{2}H} + |k-1|^{\mathbf{2}H}\right)$$

  - at lag 1: $\rho(1) = 2^{\mathbf{2}H-\mathbf{1}} - 1$; for large k: $\rho(k) \approx H(2H-1)\,k^{\mathbf{2}H-\mathbf{2}}$
  - the exponent $2H - 2 = 2d - 1$ with $d = H - 0.5$: long memory for $H > 0.5$

- Signs of the autocorrelations
  - $H > 0.5$: all autocorrelations positive (persistence); $H < 0.5$: all negative (anti-persistence); $H = 0.5$: white noise
  - **white noise**: uncorrelated values with mean 0 and constant variance

Prerequisites for Today (9/15): The ARFIMA Model

- **ARFIMA**(0,d,0) Hosking (1981) (autoregressive fractionally integrated moving average)

$$(1-L)^d X_t = \varepsilon_t, \qquad d = H - 0.5$$

  - L: the **lag operator**, $LX_t = X_{t-\mathbf{1}}$; ε_t: i.i.d. shocks; d: a fractional order of differencing
  - $d = 0$: white noise; $d = 1$: random walk; $|d| < 0.5$: stationary; $0 < d < 0.5$: long memory; $d < 0$: anti-persistence

- Binomial expansion $(1-L)^d = \sum_{j \geq \mathbf{0}} \pi_j L^j$ and its inverse $(1-L)^{-d} = \sum_{j \geq \mathbf{0}} \psi_j L^j$

$$\pi_{\mathbf{0}} = 1, \quad \pi_j = \pi_{j-\mathbf{1}} \frac{j-1-d}{j}; \qquad \psi_{\mathbf{0}} = 1, \quad \psi_j = \psi_{j-\mathbf{1}} \frac{j-1+d}{j}$$

  - π_j: AR weights, $X_t = -\sum_{j \geq \mathbf{1}} \pi_j X_{t-j} + \varepsilon_t$ depends on the whole past
  - ψ_j: MA weights, the effect of a shock from j days ago on X_t; both decay slowly

- Autocorrelations

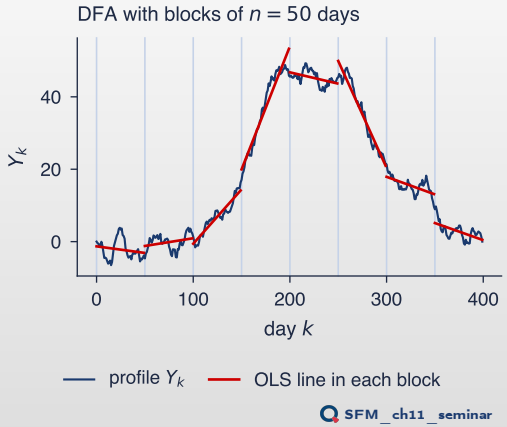
$$\rho(1) = \frac{d}{1-d}, \qquad \rho(k) = \rho(k-1)\frac{k-1+d}{k-d}$$

Prerequisites for Today (10/15): Detrended Fluctuation Analysis

- **DFA** (detrended fluctuation analysis) Peng et al. (1994)
  - 1. the **profile** $Y_k = \sum_{j \le k}(x_j - \bar{x})$, as in R/S
  - 2. cut it into blocks of size n; in each block fit a straight line $\hat{Y}_k$ by OLS (ordinary least squares)
  - 3. the **fluctuation**: the root mean square of the deviations from the lines

$$F(n) = \sqrt{\frac{1}{M}\sum_{k=1}^{M}\left(Y_k - \hat{Y}_k\right)^2} \propto n^H$$

  - M: the number of points covered by the blocks; 4. the slope of $\log F(n)$ on $\log n$ is $\hat{H}$
- Removing the local trends makes DFA less sensitive to slow changes in the level than R/S



Prerequisites for Today (11/15): The GPH Estimator

- **Periodogram** at the Fourier frequency $\lambda_j = 2\pi j/T$

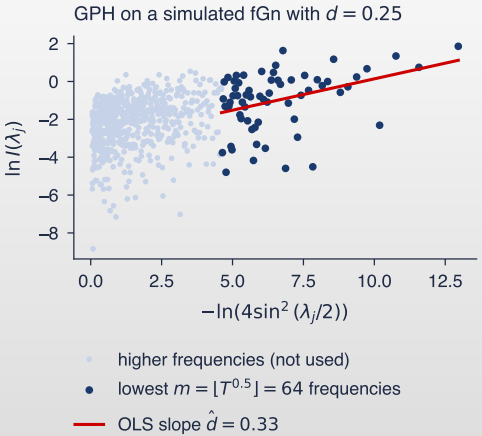
$$I(\lambda_j) = \frac{1}{2\pi T}\left|\sum_{t=1}^{T}(x_t - \bar{x})\, e^{-i\lambda_j t}\right|^2$$

  - how much of the variance comes from cycles of length T/j days; T: the number of observations; i: the imaginary unit
  - long memory: a spectrum that explodes at low frequencies, like λ^{-2d}

- **GPH** Geweke & Porter-Hudak (1983): OLS on the lowest m frequencies

$$\ln I(\lambda_j) = a + d\left[-\ln\left(4\sin^2(\lambda_j/2)\right)\right] + e_j, \quad j = 1, \ldots, m$$

  - the slope is $\hat{d}$; standard error $\pi/\sqrt{24m}$; $\hat{H} = \hat{d} + 0.5$
  - $m = \lfloor T^{0.5} \rfloor$ or $\lfloor T^{0.6} \rfloor$ ($\lfloor\cdot\rfloor$: rounding down); $m/T \to 0$ is needed



Prerequisites for Today (12/15): Lo's Modified R/S Test

- **Lo's modified R/S** Lo (1991) on all N returns

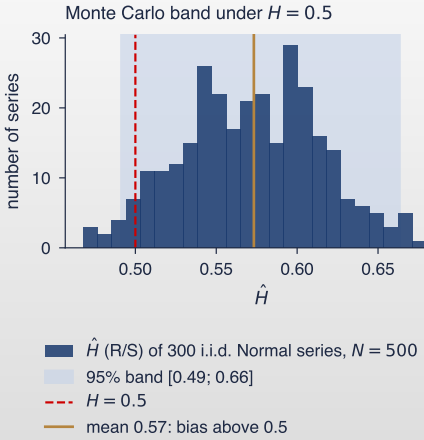
$$V(q) = \frac{R}{\sqrt{N}\,\hat{\sigma}(q)}, \qquad \hat{\sigma}^{\mathbf{2}}(q) = \hat{\gamma}_{\mathbf{0}} + 2\sum_{j=\mathbf{1}}^{q}\Big(1 - \frac{j}{q+1}\Big)\hat{\gamma}_j$$

  - R: the range of the cumulative deviations of all N returns; $\hat{\gamma}_j$: the sample autocovariance at lag j
  - $\hat{\sigma}^{\mathbf{2}}(q)$: the **Newey–West** variance Newey & West (1987) with q lags; it absorbs short memory, so only long memory remains in $V(q)$
  - q: chosen from the data (Andrews' rule); $V(0)$: the classical statistic, with $\hat{\sigma}^{\mathbf{2}}(0) = \hat{\gamma}_{\mathbf{0}}$

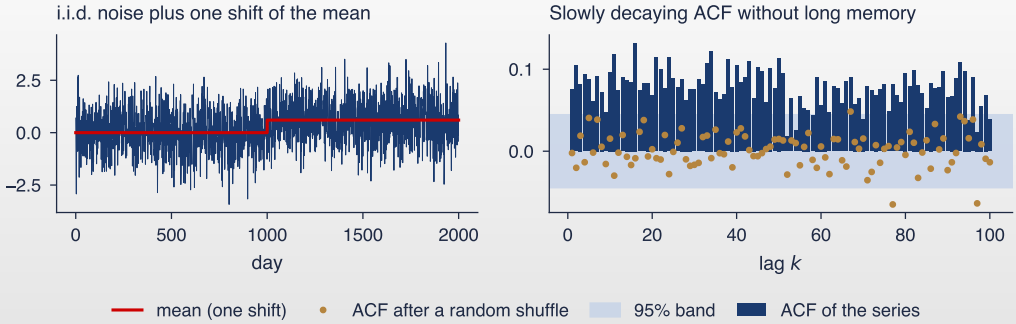
- Decision
  - $H_{\mathbf{0}}$: short memory; reject at 5% if $V(q) \notin [0.809, 1.862]$ (the 95% interval of its limit distribution)
  - $V > 1.862$: persistence; $V < 0.809$: anti-persistence

Prerequisites for Today (13/15): Monte Carlo Bands

- **Monte Carlo band** Weron (2002): what values does an estimator give when $H = 0.5$?
  - simulate many (for example 300) i.i.d. Normal series of the same length N as the data
  - apply exactly the same estimator (same block sizes or the same m) to each
  - the 2.5% and 97.5% quantiles of the simulated $\hat{H}$ form the 95% band
- Reading
  - reject $H = 0.5$ only if $\hat{H}$ lies outside the band
  - the band corrects the small-sample bias: for R/S it is not centred at 0.5
  - the shorter the series, the wider the band



Prerequisites for Today (14/15): Spurious Long Memory and the Shuffle Test



- **Spurious long memory**: slowly decaying sample ACFs without true long memory
  - causes: shifts of the mean Diebold & Inoue (2001) and persistent GARCH volatility (Chapter 9)
- **Shuffle test**: a random permutation keeps the values (the distribution) and destroys their order
  - an estimate that returns to $d = 0$ or $H = 0.5$ after shuffling depended on the order of the days

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Prerequisites for Today (15/15): Memory in Volatility and Rolling Windows

- **Memory in volatility**: the same estimators applied to $|r_t|$ and r_t^2 Ding, Granger & Engle (1993)
  - concerns the size of the returns, not their direction; the weak form of the EMH concerns the direction
  - ACF band for i.i.d. data: $\pm 1.96/\sqrt{T}$
- **Rolling windows**: $\hat{H}$ on windows of 1000 days, moved forward by 21 days, dated at the last day of the window
  - with a 5% band, about 5% of the windows fall outside it by chance
  - consecutive windows share most of their days, so their estimates are strongly correlated: only long runs outside the band matter
- Annualised volatility of a window (Chapter 8)

$$\sigma_{\text{year}} = \sqrt{A}\,\hat{\sigma}_{\text{day}}, \qquad A = 252 \text{ for stock indices}$$

A1: R/S Step by Step [Solved]

Task

- Eight daily returns (%): 0.5; -0.3; 0.8; -0.2; 0.6; -0.1; 0.4; -0.7.
- 1. Compute the mean, the deviations and the cumulative deviations $Y_1, \ldots, Y_8$.
- 2. Compute R, S (divisor n) and R/S.
- 3. S&P 500 returns since 2000 give $(R/S)_{16} = 3.91$, $(R/S)_{64} = 8.38$, $(R/S)_{256} = 17.03$; compute $\hat{H}$.
- Report: R, S, R/S, $\hat{H}$ and one sentence.

Solution

- 1. $\bar{x} = 0.125$; Y_k: 0.375, -0.050, 0.625, 0.300, 0.775, 0.550, 0.825, 0.000
- 2. $R = 0.825 - (-0.050) = 0.875$; $S = \sqrt{1.915/8} = 0.489$; $R/S = 1.788$
- 3. $\log_{10}$: n: 1.204, 1.806, 2.408; R/S: 0.592, 0.923, 1.231; $\hat{H} = 0.639/1.204 = 0.531$
- Alternating signs keep the cumulative deviations small; $\hat{H}$ slightly above 0.5 is the small-sample bias of R/S, not memory (B1).

Back Exercise list

A2: Grouped Signs [Proposed]

Solution discussed in class

Task

- Eight returns (%): 0.6; 0.4; 0.5; 0.3; −0.4; −0.6; −0.2; −0.5. Model: A1.
- 1. Compute the mean and $Y_1, \ldots, Y_8$.
- 2. Compute R, S and R/S, and compare with A1.
- 3. BET returns give $(R/S)_{16} = 4.23$, $(R/S)_{64} = 10.08$, $(R/S)_{256} = 21.61$; compute $\hat{H}$.
- Report: R/S, $\hat{H}$ and two sentences.

Back Exercise list

A3: Fractional Gaussian Noise and 10-Day Risk [Solved]

Task

- Daily returns follow an fGn with $H = 0.7$ and daily volatility $\sigma = 1.2\%$; $z_{0.01} = -2.326$.
- 1. Compute $\rho(1)$ and $\rho(10)$ exactly and $\rho(10)$ with the approximation $H(2H-1)k^{2H-2}$.
- 2. Compute the scaling factor 10^H and compare it with $\sqrt{10}$.
- 3. Compute the one-day and the 10-day VaR 1% (Normal) with h^H and with $\sqrt{h}$.
- 4. Compute the fractal dimension of the price path.
- Report: six numbers and one sentence.

Solution

- 1. $\rho(1) = 2^{0.4} - 1 = 0.320$;
  $\rho(10) = \frac{1}{2}(11^{1.4} - 2 \cdot 10^{1.4} + 9^{1.4}) = 0.0704$;
  approximation $0.7 \cdot 0.4 \cdot 10^{-0.6} = 0.0703$
- 2. $10^{0.7} = 5.012$ against $\sqrt{10} = 3.162$
- 3. $2.326 \times 1.2 = 2.79\%$; 10 days: $2.79 \times 5.012 = 13.99\%$ against 8.83% ($+58.5\%$)
- 4. $D = 2 - H = 1.30$
- Persistent returns make multi-day losses add up: the $\sqrt{h}$ rule understates 10-day risk by more than a third.

Back Exercise list

A4: Anti-Persistence and a Small Deviation from 0.5 [Proposed]

Solution discussed in class

Task

- The setting of A3 with $H = 0.4$ and with $H = 0.55$. Model: A3.
- 1. Compute $\rho(1)$ and $\rho(10)$ for both values of H.
- 2. Compute the 10-day VaR 1% for both and its difference from the $\sqrt{h}$ rule, in %.
- 3. Compute D for both.
- Report: eight numbers and one sentence.

A5: ARFIMA(0, 0.3, 0) against an AR(1) [Solved]

Task

- $(1-L)^{\mathbf{0.3}} X_t = \varepsilon_t$, ε_t i.i.d. $N(0,1)$.
- 1. Compute $\pi_{\mathbf{1}}$, $\pi_{\mathbf{2}}$ and the MA weights $\psi_{\mathbf{1}}$, $\psi_{\mathbf{2}}$.
- 2. Compute $\rho(1)$, $\rho(2)$, $\rho(10)$ and $\rho(50)$.
- 3. For an AR(1) with $\phi = \rho(1)$, compute $\rho(10)$ and $\rho(50)$.
- 4. Give H and the first lag with $\rho(k) < 0.05$ for both models.
- Report: ten numbers and one sentence.

Solution

- 1. $\pi_{\mathbf{1}} = -0.3$, $\pi_{\mathbf{2}} = -0.3 \times 0.7/2 = -0.105$; $\psi_{\mathbf{1}} = 0.3$, $\psi_{\mathbf{2}} = 0.3 \times 1.3/2 = 0.195$
- 2. $\rho(1) = 0.3/0.7 = 0.429$; $\rho(2) = 0.429 \times 1.3/1.7 = 0.328$; $\rho(10) = 0.173$; $\rho(50) = 0.091$
- 3. $0.429^{\mathbf{10}} = 0.0002$; $0.429^{\mathbf{50}} \approx 0$
- 4. $H = 0.8$; below 0.05 after 4 lags (AR) and 222 lags (ARFIMA)
- Same $\rho(1)$, very different memory: the AR(1) forgets in days, the ARFIMA in months.

Back Exercise list

A6: Strong Memory and Anti-Persistence [Proposed]

Solution discussed in class

Task

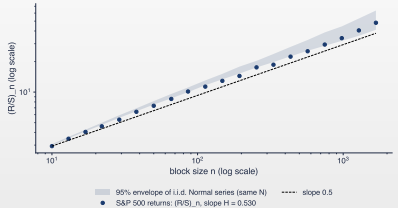
- ARFIMA(0,d,0) with $d = 0.45$ and with $d = -0.2$. Model: A5.
- 1. Compute π_1, π_2, $\rho(1)$, $\rho(2)$ and $\rho(10)$ for both.
- 2. For $d = 0.45$, compare $\rho(10)$ and $\rho(50)$ with an AR(1) with $\phi = \rho(1)$.
- 3. Give H for both and say which one is stationary.
- Report: a small table and one sentence.

Back Exercise list

B1: Long Memory in S&P 500 Returns? [Solved]

- **Question**: do daily S&P 500 returns have long memory?
- **Inputs**: S&P 500 closes since 2000, daily log returns in %
- **Tasks**
  1. Estimate H with R/S and DFA (block sizes from 10 to $N/4$) and d with GPH ($m = \lfloor T^{0.5} \rfloor$), with the standard error of $\hat{d}$.
  2. Compute Lo's modified statistic $V(q)$ with Andrews' q, and the classical statistic $V(0)$.
  3. Simulate 300 i.i.d. Normal series of the same length and give the 95% band of each estimator.
  4. Draw the R/S plot inside the Monte Carlo envelope.
  5. Interpretation: is the R/S estimate above 0.5 evidence of long memory?
- **Report**: a table of four estimates with their bands, the chart and two sentences
- Notebook: section B1

B1: Solution [Solved]



	R/S	DFA	GPH $\hat{H} = \hat{d} + 0.5$	Lo $V(q)$
estimate	0.530	0.488	0.536	1.559 ($q = 7$)
95% band	[0.512, 0.575]	[0.456, 0.540]	[0.363, 0.652]	[0.809, 1.862]

- $N = 6\,714$; GPH: $\hat{d} = 0.036$, SE 0.071 ($m = 81$), $t = 0.51$; classical $V(0) = 1.379$
- Interpretation: no: i.i.d. series of this length give R/S 0.544 on average; every estimate lies inside its band and Lo does not reject short memory

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B2: BET and Banca Transilvania [Proposed]

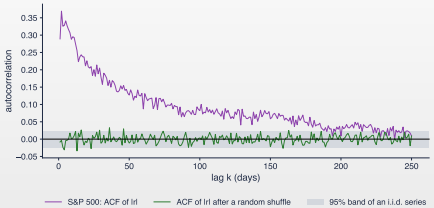
- **Question**: does the BET or Banca Transilvania show long memory in returns? Model: B1.
- **Inputs**: BET since 2000, TLV since 2010 (without 30–31 May 2016); daily log returns in %
- **Tasks**
  1. Estimate H with R/S, DFA and GPH for each series.
  2. Compute Lo's $V(q)$ and say whether short memory is rejected at 5%.
  3. Compute the Monte Carlo band of each estimator for each length.
  4. Interpretation: what could explain the result for the BET?
- **Report**: one table and two sentences
- Notebook: section B2

B3: Long Memory in S&P 500 Volatility [Solved]

- **Question**: do the absolute and the squared S&P 500 returns have long memory?
- **Inputs**: S&P 500 daily log returns since 2000; $|r_t|$ and r_t^2
- **Tasks**
  1. Estimate d with GPH for $m = \lfloor T^{0.5} \rfloor$ and $m = \lfloor T^{0.6} \rfloor$, and H with DFA, for $|r_t|$ and r_t^2.
  2. Report the ACF of $|r_t|$ at lags 1, 50 and 250 and the 95% band $\pm 1.96/\sqrt{T}$.
  3. Shuffle $|r_t|$ at random and repeat the estimates.
  4. Draw the ACF of $|r_t|$ before and after the shuffle.
  5. Interpretation: does long memory in $|r_t|$ contradict the weak form of the EMH?
- **Report**: a table, the chart and two sentences
- Notebook: section B3

Back Exercise list

B3: Solution [Solved]



	GPH $T^{0.5}$	GPH $T^{0.6}$	DFA	$\rho(1)$	$\rho(50)$	$\rho(250)$
$\lvert r_t \rvert$	0.470	0.467	0.953	0.289	0.127	0.009
r_t^2	0.326	0.377	0.892	0.313	0.053	−0.010
$\lvert r_t \rvert$ shuffled	0.028	−0.001	0.498	−0.008	−0.011	0.009

- $m = 81$ and 197; SE 0.071 and 0.046; ACF band ± 0.024; after the shuffle all estimates fall to 0 (d) or 0.5 (H)
- Interpretation: no: the weak form concerns the direction of returns (B1); long memory in $|r_t|$ means predictable risk, which needs a volatility model (Chapter 9)

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B4: Bitcoin Returns and Volatility [Proposed]

- **Question**: does Bitcoin show long memory in returns, in volatility, or in both? Models: B1, B3.
- **Inputs**: Bitcoin daily log returns since 2014 (7 days a week)
- **Tasks**
  1. Estimate R/S, DFA, GPH and Lo's $V(q)$ for the returns, with the Monte Carlo bands.
  2. Estimate GPH ($T^{0.5}$, $T^{0.6}$) and DFA for $|r_t|$ and r_t^2.
  3. Repeat the volatility estimates after a random shuffle of $|r_t|$.
  4. Interpretation: is Bitcoin closer to the S&P 500 or to the BET?
- **Report**: one table and two sentences
- Notebook: section B4

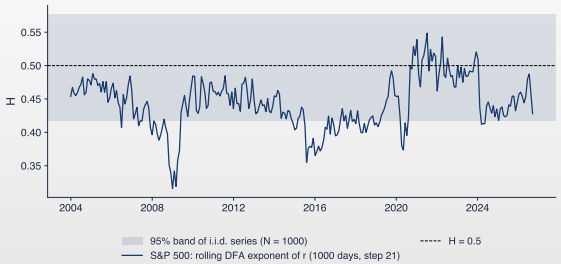
Back Exercise list

B5: Rolling Hurst Exponents of the S&P 500 [Solved]

- **Question**: has the memory of S&P 500 returns changed since 2000?
- **Inputs**: S&P 500 daily log returns since 2000
- **Tasks**
  1. Estimate the DFA exponent on windows of 1000 days moved by 21 days, dated at the last day of each window.
  2. Compute the 95% Monte Carlo band for $N = 1000$.
  3. Report the minimum and the maximum with their dates, and the share of windows below and above the band.
  4. Draw the rolling exponent with the band.
  5. Interpretation: does a window outside the band prove that the market was inefficient at that time?
- **Report**: six numbers, the chart and two sentences
- Notebook: section B5

Back Exercise list

B5: Solution [Solved]



- 273 windows from 29 December 2003; band $[0.417, 0.577]$; mean $\hat{H} = 0.443$
- Minimum 0.316 (31 December 2008), maximum 0.549 (9 July 2021); 22.3% of the windows below the band, 0.0% above
- Interpretation: no: by chance about 14 of 273 windows fall outside, and consecutive windows overlap; only long runs count; here the runs are below the band (anti-persistence after 2008 and in 2016), never above

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B6: Rolling Exponents of the DAX and of Banca Transilvania [Proposed]

- **Question**: does the result of B5 hold for the DAX and for Banca Transilvania? Model: B5.
- **Inputs**: DAX since 2000, TLV since 2010; daily log returns in %
- **Tasks**
  1. Estimate the rolling DFA exponent as in B5.
  2. Report the share of windows below and above the band, and the minimum and maximum with their dates.
  3. Compare the mean exponent in the first and in the last third of the windows.
  4. Interpretation: which market looks more stable over time?
- **Report**: one table and two sentences
- Notebook: section B6

Back Exercise list

C1: Does Market Memory Change in Crises? [Proposed]

- **Question**: does the Hurst exponent of returns or of volatility change in a crisis, as the fractal market hypothesis suggests?
- **Inputs**: S&P 500 and BET daily log returns; a calm year (2017) and two crisis years (September 2008–August 2009, February 2020–February 2021); models: B1, B3
- **Tasks**
  1. Estimate the DFA exponent of r_t and of $|r_t|$ and the annualised volatility in each of the three years, for both indices.
  2. Compute the Monte Carlo band for windows of about 250 days.
  3. List three reasons why one crisis year cannot settle the question.
  4. Interpretation: which design would separate a change of memory from a change of volatility?
- **Report**: a table and a plan for a project
- Notebook: section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to interpret long-memory estimates for the S&P 500 (daily, since 2000). The answer:
- (a) R/S gives H = 0.530 for the returns: above 0.5, so returns have long memory and trends can be exploited.
- (b) H = 0.530 means a 53% probability that tomorrow has the same sign as today.
- (c) GPH gives d = 0.470 for |r|, so H = d - 0.5 = −0.030: absolute returns are anti-persistent.
- (d) Long memory in |r| contradicts the weak form of the EMH.
- (e) Lo's modified statistic V = 1.56 lies inside [0.809, 1.862], so short memory is not rejected at 5%.
- (f) For more precision, use all T/2 = 3357 frequencies in the GPH regression.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

Points to Remember

- R/S, DFA and GPH estimate the same H in different ways; report several, each with a Monte Carlo band for the same N
- $d = H - 0.5$; H is a scaling exponent, not a probability
- Returns of large markets: no long memory; volatility: long memory everywhere; the shuffle test locates it in the order of the days
- Rolling estimates: read long runs outside the band, not single windows
- An AI answer is a draft: check the band, the conversion between d and H, and the bandwidth

After the Seminar

- Lecture 11 develops each topic of today: the FMH, self-similarity, fBm and ARFIMA, the estimators, spurious memory, rolling exponents, risk scaling
- C1 can grow into a team project: GARCH-standardised returns, many crises, a GARCH-based band
- Reading: Franke, Härdle & Hafner (2019), Ch. 14; Lo (1991); Weron (2002)

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